



FLORIDA STATE UNIVERSITY
DEPARTMENT OF SCIENTIFIC COMPUTING

WRITTEN PRELIMINARY EXAMINATION
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Question - 9

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1 Statistics[Dr. Beerli]

1.1 Part 1

To get an idea of the growth model to use, I first plotted the time series of bacterial growth. Before this I first asked ChatGpt(conversation is [attached](#)) about the potential growth models which can be used and under what cases can they be used. The different growth models which are commonly used for bacterial growth are:

1. **Exponential Growth Model:** Appropriate when bacterial growth is in exponential phase.
2. **Gompertz Growth Model:** Appropriate when bacterial growth shows a sigmoidal growth pattern.
3. **Monod Model:** Appropriate when the growth rate depends on the concentration of a limiting nutrient.
4. **Logistic Growth Model:** Appropriate when the growth rate has a saturation point.

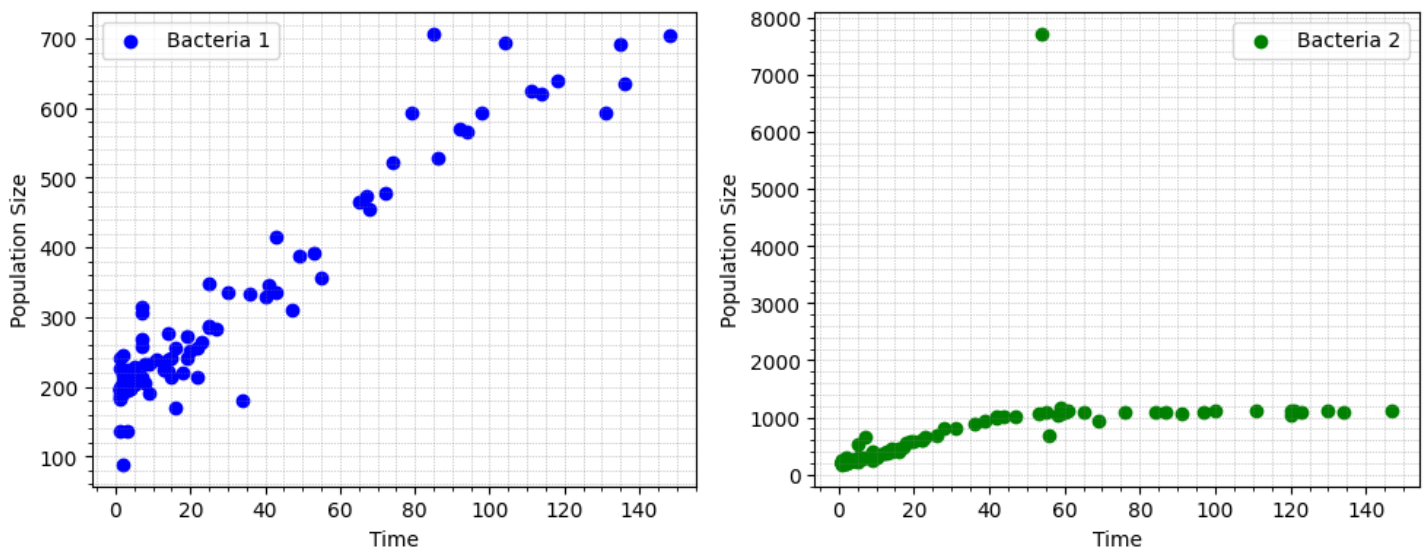


Figure 1: Plots of given data as a function of time.

As clear from the Fig-1, the data is varying logarithmically and hence Logistic Growth model should be used. Another reason for choosing Logistic Growth model is that the plots for sure doesn't show exponential growth behaviour and I am not familiar with the remaining two models. For simplicity, I am taking two parameters only in this model. This was also suggested by ChatGpt.

Logistic Growth model:

The logistic growth model describes how a population grows over time in an environment with limited resources. It is a commonly used model in ecology to represent the growth of populations, including bacterial growth. The model is characterized by an initial exponential growth phase, followed by a slowdown as the population approaches the carrying capacity of the environment.

$$N(t) = \frac{K}{1 + \left(\frac{k - N_0}{N_0} \right) e^{-rt}}$$

where:

- $N(t)$ is the population size at time t .
- N_0 is the initial population size.
- K is the carrying capacity (parameter 1).
- r is the intrinsic growth rate (parameter 2).

1.2 Part 2

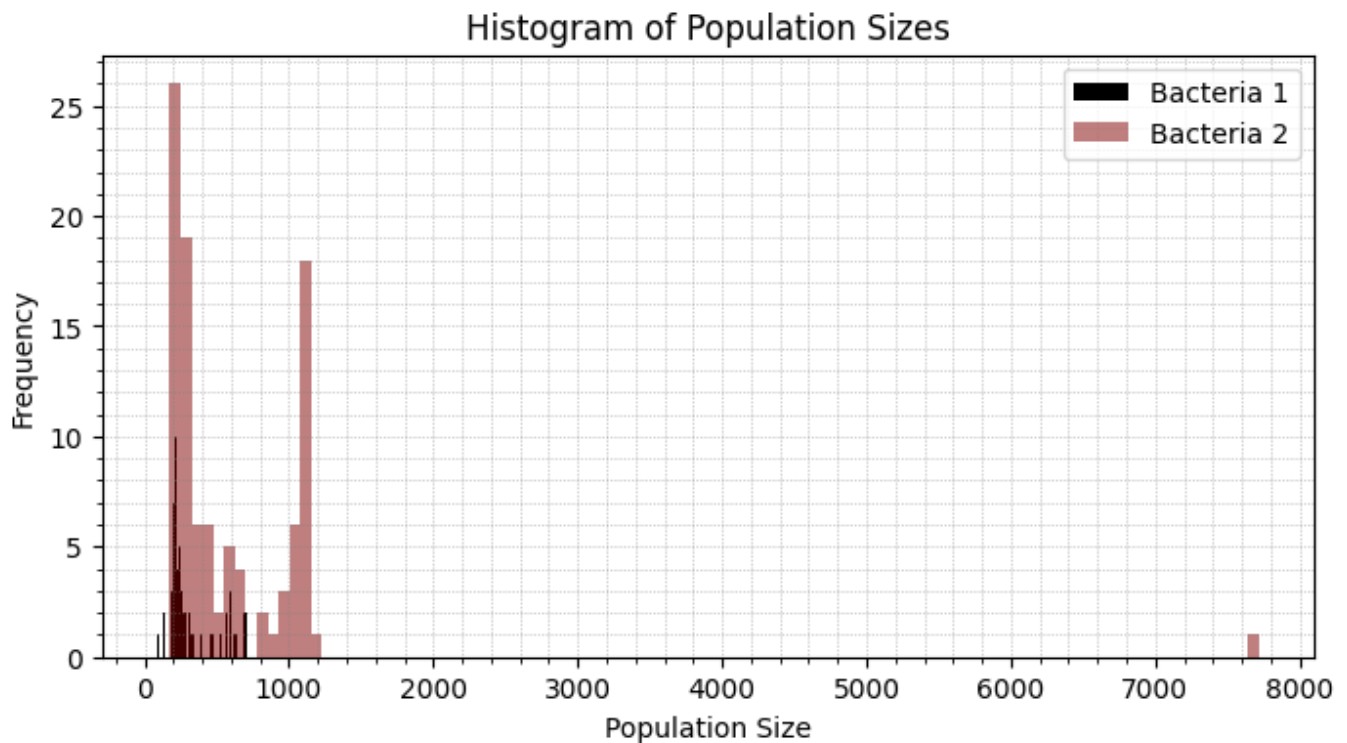


Figure 2: Histogram of both bacteria for comparison.

As clear from the Fig-2 above, the growth of bacteria 2 is more spread out and has higher population size compared to bacteria 1 which is having population size at a narrower range. The population of bacteria 1 stays in range of 100 to 700. Whereas the population of bacteria ranges approximately between 200 to 1200. One more interesting point is the sudden jump in population of bacteria 2 at $t = 54$. The population suddenly becomes around 7000 which is also shown in the histogram. I don't know if this sudden jump in population is possible in reality or not. This may be because some error in data collection.

1.3 Part 3

I have used Wikipedia page [\[2\]](#) and Youtube [video](#) to study about Metropolis-hastings scheme. ChatGpt helped me to debug my code.

Metropolis Hastings Algorithm:

```

1 Input:
2 - Initial state x_0
3 - Proposal distribution q(y|x)

```

```

4 - Target distribution p(x)
5 - Number of iterations
6
7 Output:
8 - Samples from the target distribution
9
10 Initialize the Markov chain at an arbitrary state x_0.
11 For each iteration t = 1, 2, ...:
12     Generate a proposal state y from the proposal distribution q(y|x_{t-1}).
13     Calculate the acceptance probability:
14         alpha = min(1, p(y) * q(x_{t-1}|y) / (p(x_{t-1}) * q(y|x_{t-1})))
15     Generate a uniform random number u from the interval [0, 1].
16     If u <= alpha, accept the proposal and set x_t = y; otherwise, reject
        the proposal and set x_t = x_{t-1}.
17 Repeat the above steps for a sufficient number of iterations to obtain a
        representative sample from the target distribution.

```

I varied my initial parameters and iterations until I got results in the credible intervals. For confidence interval of 95%, I got the following results with 1000000 iterations in the algorithm. For the remaining question I used $k = 1000$ and $r = 0.1$ as initial guess of my parameters.

Results:

```

=====
Posterior Summary Statistics:
-----
Bacteria 1:
K: Mean = 740.64, Std = 133.05, Credible Interval = [626.15, 985.07]
r: Mean = 0.04, Std = 0.01, Credible Interval = [0.03, 0.05]
-----
Bacteria 1:
K: Mean = 1279.1786, Std = 27.0274, Credible Interval = [1181.4383, 1285.6450]
r: Mean = 0.1197, Std = 0.0019, Credible Interval = [0.1192, 0.1264]
=====

```

Once, I got the final parameters within credible intervals, I plotted the results with raw data. Following are the results I am getting. They are within the credible intervals defined. Fig-3 shows the evolution of k and r for 10^6 iterations. For calculation of final statistics first 10^3 iterations are discarded.

```

-----
Parameters for Bacteria 1:
-----
Mean K: 740.635076632971
Mean r: 0.041418087653550005
Final K: 626.5084346930691
Final r: 0.049222993499177116
-----
Parameters for Bacteria 2:
-----
Mean K: 1279.1786095887194
Mean r: 0.1197049948309464
Final K: 1284.8533881381127
Final r: 0.1193281096824759

```

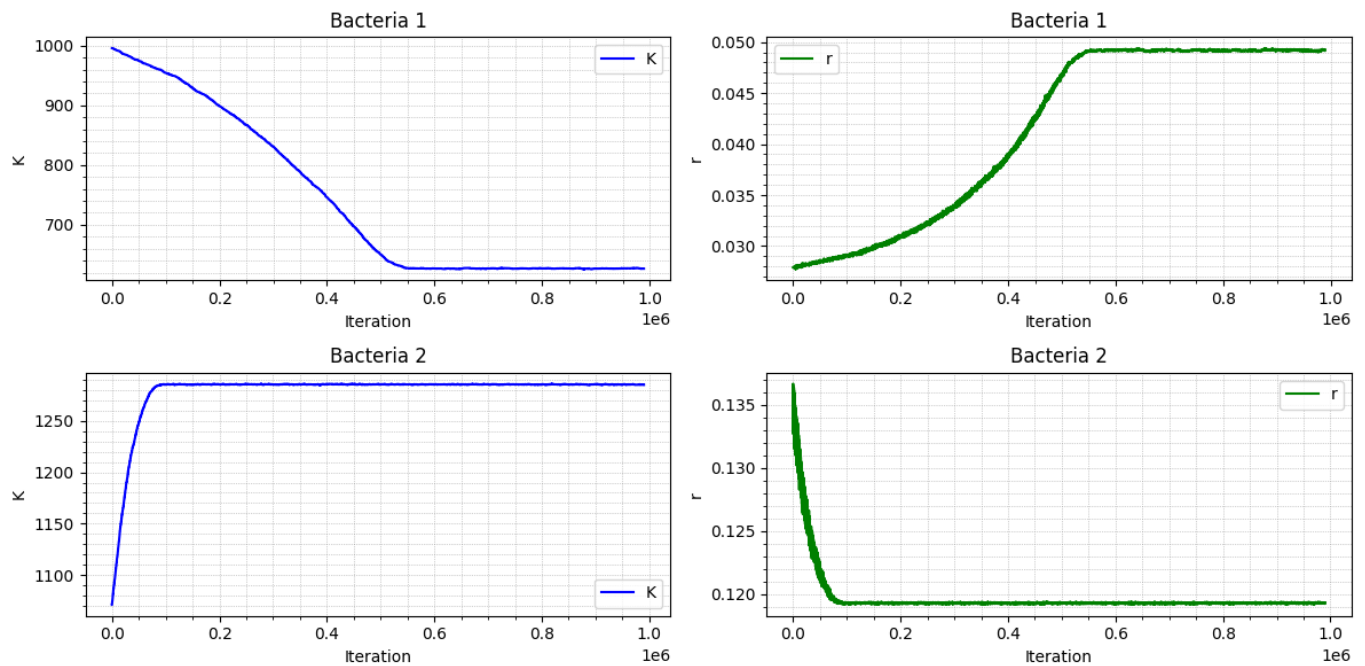


Figure 3: Growth of Parameters K and r with iteration for Metropolis Hastings Algorithm.

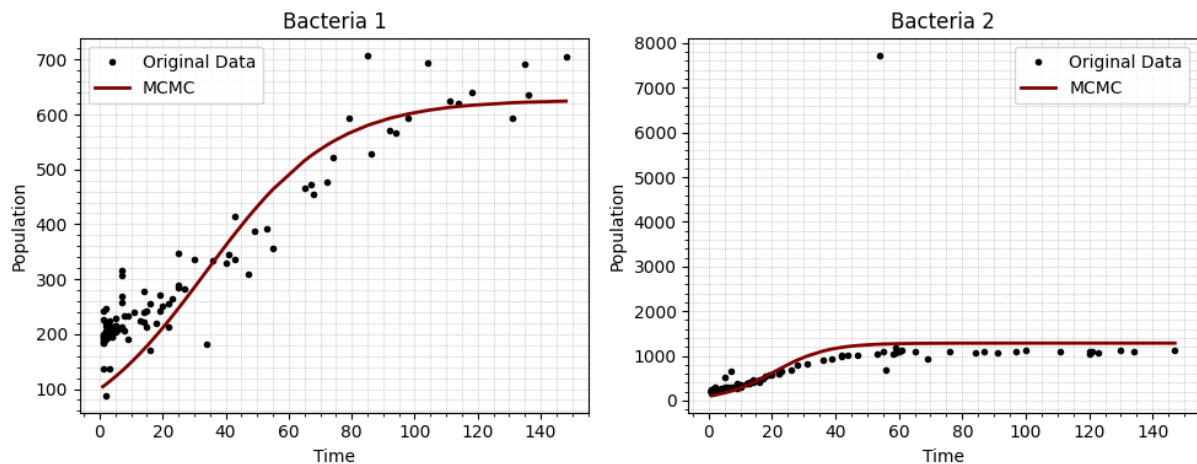


Figure 4: Comparison of obtained model with parameters estimated using Metropolis-Hastings and raw data from the two datasets- bacteria1.txt and bacteria2.txt

1.4 Part 4

I have tried using two methods. Initially, I was trying to implement Expectation-minimization algorithm, but due to some issues it was underestimating parameters and the values were not within credible intervals as well. I was not able to figure out issues even with the help of AI. So, I implemented Particle Swarm Optimization. I used the Wikipedia page [3] for Particle Swarm Optimization to read about it and I took the code from [tutorial](#) given on GeeksforGeeks[1] .

Particle Swarm Optimization Algorithm:

```

1 Input:
2   - t: Time points
3   - population: Observed population sizes
4   - num_particles: Number of particles (default: 30)

```

```

5 - max_iter: Maximum iterations (default: 10000)
6 - w: Inertia weight (default: 0.5)
7 - c1: Cognitive parameter (default: 1)
8 - c2: Social parameter (default: 2)
9
10 Output:
11 - Optimal parameters (K, r)
12
13 Initialize swarm with random positions and velocities
14 Initialize personal and global best positions and scores
15
16 For each iteration:
17     For each particle:
18         Update velocity:
19             velocity = w * velocity + c1 * rand() * (personal_best -
20 position) + c2 * rand() * (global_best - position)
21         Update position: position += velocity
22         Evaluate fitness: mean squared error (MSE)
23         Update personal best if fitness improves
24         Update global best if any personal best improves
25
26 Return global best position (optimal K and r)

```

Results:

Final K and r values for Bacteria 1:

K: 626.7679919860823

r: 0.049196367822090185

Final K and r values for Bacteria 2:

K: 1285.298819322784

r: 0.11928223995070214

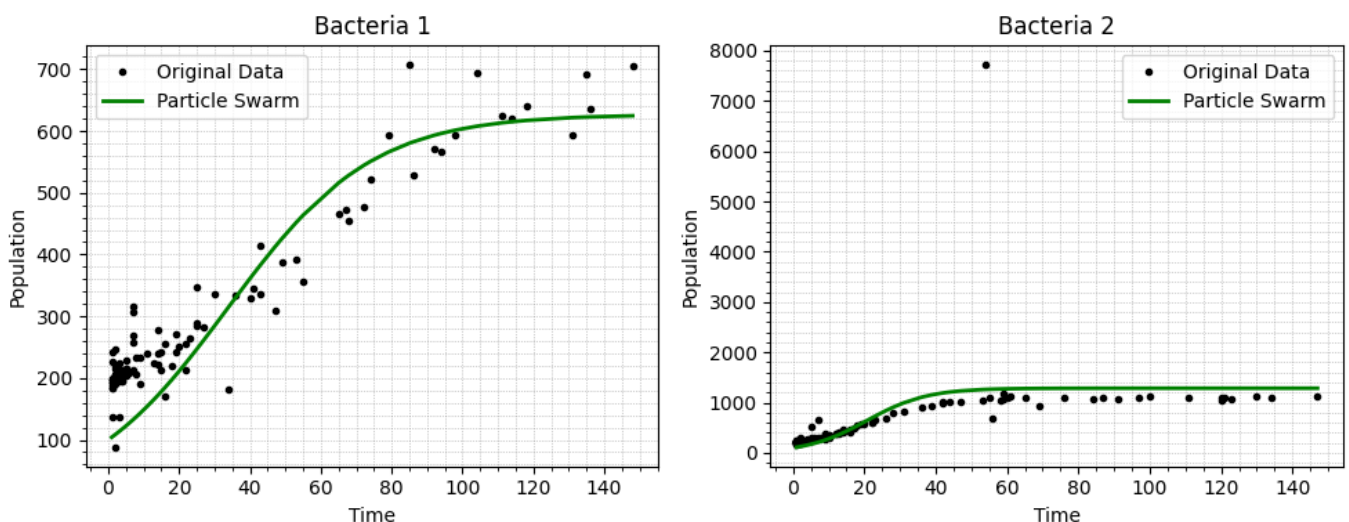


Figure 5: Comparison of obtained model with parameters estimated using the three algorithms and raw data from the two datasets- bacteria1.txt and bacteria2.txt

1.5 Part 5

The parameters estimated using Particle swarm optimization and Metropolis-Hastings are almost same suggesting that the model works well. Also, these parameters lie within the support interval. They fit the model very well which can be observed from Fig-6. However, there is some issue with Expectation minimization method and it is not able to generate good results. Table-1 compares the parameter obtained using three methods and also gives the support intervals.

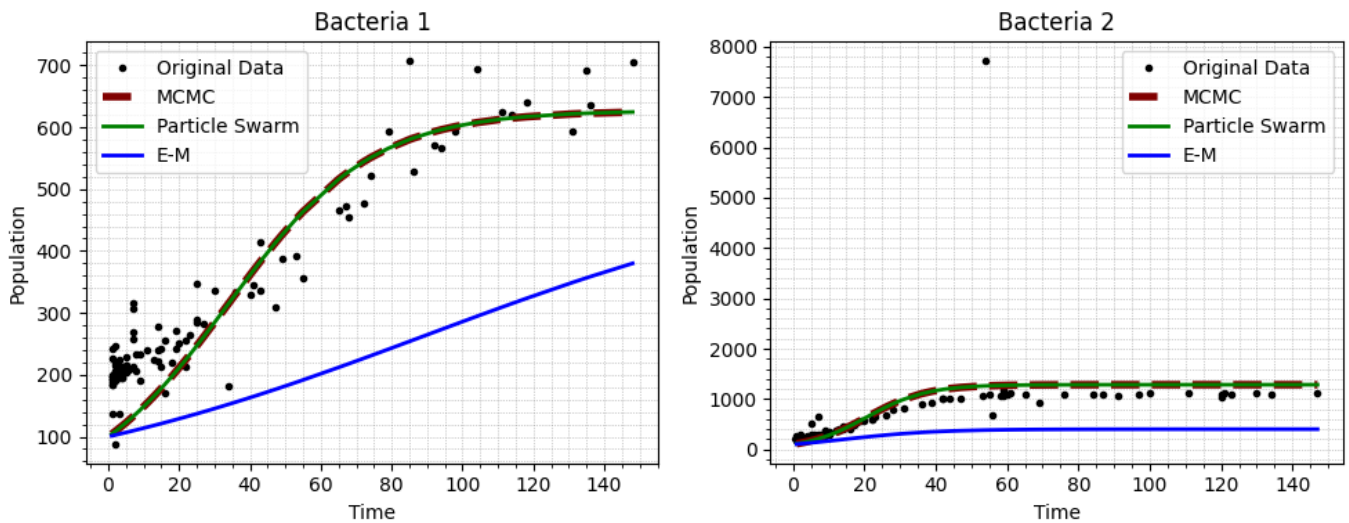


Figure 6: Comparison of obtained model with parameters estimated using Particle Swarm Optimization and raw data from the two datasets- bacteria1.txt and bacteria2.txt

Bacteria 1				
Parameter	MCMC	PSO	EM	Support Interval
K	626.51	626.77	527.50	(626.15, 985.07)
r	0.05	0.05	0.02	(0.0300, 0.0500)
Bacteria 2				
Parameter	MCMC	PSO	EM	Support Interval
K	1284.78	1285.30	401.86	(1181.4383, 1285.6450)
r	0.12	0.12	0.08	(0.1192, 0.1264)

Table 1: Parameter values for Bacteria 1 and Bacteria 2 from MCMC, PSO, and EM methods and credible intervals.

The results here suggests that the logistic Growth model adopted at the beginning was correct.

References

- [1] GeeksforGeeks. Particle swarm optimization (pso) – an overview, Year not specified.
- [2] Wikipedia contributors. Metropolis-hastings algorithm. https://en.wikipedia.org/wiki/Metropolis%E2%80%93Hastings_algorithm.
- [3] Wikipedia contributors. Particle swarm optimization. https://en.wikipedia.org/wiki/Particle_swarm_optimization.

[4] ChatGpt conversation: <https://chatgpt.com/share/81c491ac-536e-452d-a08c-2adf6d260d20>

[5] YouTube: https://www.youtube.com/watch?si=Sp7OnRXv1PiVzHKb&v=MxI78mpq_44&feature=youtu.be